Nikhil Nair

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Summary

Software engineer with hands-on experience building edge CV pipelines, robotics perception systems, and hardware-software integrations. Currently leading R&D at Super FabLab Kerala, with a published patent, 4 research publications, and a B.Tech in Computer Science & Engineering. Open to embedded, SDE, or AI/ML engineering roles.

Experience

Super FabLab Kerala – MIT & KSUM — **Nov 2025 – Present**
Software Development Engineer - Technology Fellow — *Kochi, India*

- Reduced CNC flatbed costs and cut overall fabrication time, achieved by eliminating the need for expensive onboard cameras, by developing a lightweight **JavaScript** and **ArUco marker** vision pipeline that translates multicolor sketches into machine commands via smartphone cameras.
- Cut manual setup overhead by **80%** by implementing a **computer vision pipeline** for automated bed calibration; achieved **0.2mm alignment precision** via real-time object localization on an embedded platform.
- Reduced unit cost to **INR 2,000** (vs.∼ INR 40,000 industrial alternatives) – a **95% cost reduction**, saving **INR 10L+** at scale, by engineering a custom **ESP32 Web-Plotter** with client-side G-code parsing.

U&I Trust — **Aug 2024 – Apr 2025**
Volunteer Tutor — *Mumbai, India*

- Led academic sessions for underprivileged students; supported a INR 2.13 crore education fundraiser through outreach.

TATA Elxsi — **Oct 2023 – Dec 2023**
AI and Robotics Intern (Transport Business Unit) — *Chennai, India*

- Improved obstacle detection by fusing 3D LiDAR with 2D vision using **Python**, extending range in tough conditions.
- Optimized KITTI dataset preprocessing by ∼30% using **ROS2** and **OpenCV**, enhancing the perception pipeline using RANSAC.

Education

Vellore Institute of Technology - Chennai — **Sept 2021 - Sept 2025**
B.Tech in Computer Science and Engineering — *CGPA: 8.21/10*

- **Relevant Coursework:** DSA, SDLC, Operating Systems, Computer Networks, Artificial Intelligence, Control Systems

Projects

Urumi V2 — Open Source Camera-Assisted 4 Axis CNC Cutter — **MIT Collaboration | Jan 2026 – Present**

- Implementing camera integration, sensor fusion, and lightweight UI architecture for this MIT-collaborated CNC project.
- Developing algorithms for **camera calibration** and automated material alignment; presented project architecture and vision-guided cutting capabilities at **MIT FAB26 @ Boston**.

NAS-Optimized Deep Learning Model for Concrete Strength Prediction — **Thesis | 2025**

- Engineered a **Neural Architecture Search**-optimized model in **Python** & **Keras** for concrete strength prediction; improved accuracy by ∼10% ($R^2 > 0.90$). Supported design of stronger, cost-effective, and sustainable mixes.

Doki Doki — Restaurant Pager with Gaming Console — **Jul 2026 – Aug 2026**

- Built a retro gaming restaurant pager using XIAO ESP32-C3, custom PCB, and 3D-printed enclosure for Seeed Studio Interactive Signage Contest 2026; boosts sales via ads and streamlined ordering and frictionless payment pipeline.

Technical Skills

Languages: Java, Python, C, SQL, PHP, HTML/CSS, JavaScript, React.js, Node.js
Frameworks/Tech: Spring Boot, Docker, Flask, DOM manipulation, TensorFlow, REST API, OpenCV, ROS2, Jenkins
Tools/Platforms: Git, Google Cloud Platform, Arduino, KiCAD, Fusion360, Figma, CI/CD, Gemini, Claude
Soft Skills: Communication, Team Collaboration, Problem Solving, Presentation, Analytical Thinking, Creativity

Patent & Publication

A System to Recognize Number Plate and Allow Entry of a Vehicle — **2024**
Patent Application Number: 202441051527 — *Published*

Comprehensive Review of Blockchain-Oriented Methods in Agricultural SCM — **2026**
Harvesting Data: Blockchain, AI and Advanced Innovations in Agriculture — *Accepted*

Inverter-Integrated ECC w/ Torsion Points as Black Box Alternatives in Aircraft Security — **2025**
Soft Computing Applications in Modern Power and Energy Systems — *Accepted*

Achievements & Certifications

Achievements: Invited to Talk at **IndiaFOSS 2026**; Awarded grant for Top 50 ideas by DigiKey. Awarded **PMSS Scholarship** for academic excellence & leadership; **Open source contributor** - merged PRs in projects with **1K+ stars.**
Certifications: Quantum Computing (Udemy, 2025) – covered **Python**, **Q#**, **Qiskit**; ML Specialization (Coursera, 2023); **Google Cloud** Career Practitioner & Foundation Programs (2023–24).